



When Your Cloud Goes Down, Your Business Doesn't

**Portable Snapshots. Tested Cross-Cloud Failover. No
Egress Tax.**

A technical brief on cloud-portable disaster recovery

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The Problem in One Page

On October 20, 2025, an automated system deleted a single DynamoDB DNS record in AWS US-EAST-1. **141 AWS services went dark for roughly 15 hours**. Over 1,000 companies, including Snapchat, Robinhood, Coinbase, Uber, and Delta, lost both production and disaster recovery at the same time, because every one of them ran AWS-native DR (AWS Post-Event Summary; Downdetector logged 16M+ problem reports).

Eight days later, an unintentional config change crashed Azure Front Door nodes across the **entire global fleet for 9 hours**, taking down Microsoft 365, Xbox, Azure AD B2C, Databricks, SQL Database, and Scottish Parliament voting.

Critical outages at the Big Three increased **18% YoY in 2024**, lasted **18.7% longer**, and totaled nearly 100 hours across six incidents each exceeding 10 hours (Parametrix 2024). The Global 2000 lost a collective **\$400B annually**, roughly \$200M per company, to downtime (Splunk/Oxford Economics 2024).

Yet **89% of organizations claim a multi-cloud strategy** (Flexera 2024) while only **13% have orchestrated DR workflows that actually run across clouds** (Veeam 2024 Data Protection Trends). That 76-point gap is what this brief is about: why cloud-native DR fails when the cloud fails, and what the architectural alternative looks like.

Next thing to think about: when you last ran a recovery to a cloud you are not currently running in.

The Cloud Outage Reality

Two incidents, one architectural lesson



AWS US-EAST-1, October 20, 2025. A rare timing race between redundant DNS components let an automated system delete the DynamoDB regional endpoint record. 141 services failed for ~15 hours. AWS Elastic Disaster Recovery customers could not fail over, because EDR's only recovery target is AWS itself (AWS Post-Event Summary; NBC News; CNN).

Azure Front Door, October 29, 2025. An unintentional configuration change propagated to every Azure Front Door node in the global fleet at once and crashed them for 9 hours. The blast radius was the entire Azure edge: Azure AD B2C, Databricks, SQL Database, Microsoft 365, Xbox, Minecraft, Costco, Starbucks, Alaska Airlines, and Scottish Parliament voting all went down simultaneously (Microsoft Post-Incident Review; The Register).

The Front Door failure is the more instructive of the two, because Front Door is the canonical reference for "Azure-native HA": Microsoft's own documentation recommends it as the global load-balancer for highly available workloads. Regional failover **within** Azure did nothing on October 29, because the failure was not regional. The control plane that selects between regions was itself the failure, and there is no in-Azure architecture that routes around the Azure global control plane.

Workloads behind Front Door with multi-region Azure backends, paired regional storage, and Azure Site Recovery as the DR tier still went dark for 9 hours. The DR estate inherited the production estate's substrate, so it inherited the production estate's outage.

The common factor with US-EAST-1: each tool that recovers "to the same cloud you backed up from" recovered to nothing on the day the cloud itself was the failure.

Next thing to think about: in each of the three incidents above, where would your runbook have failed over to, and have you tested that path this quarter?

The Economics of Downtime

The threat is the operator, not the workload

Human error caused 68% of all cloud service interruptions in 2024 (Parametrix). That single number reframes the entire DR planning conversation. The implicit assumption behind cloud-native DR (the cloud itself will hold; only your workload will fail) is wrong roughly two times out of three. The operator with privileged write access to the cloud is the more probable failure mode than the cloud's hardware or software.

Two recent zone-level examples confirm the operator-error dominance. Google Cloud's April 2023 Paris data center suffered a water intrusion event that took the zone offline for **two weeks** (Google Cloud Service Health post; Reuters coverage). Google Cloud's June 12, 2025 global Service Control quota-system push contained a malformed config and broke dozens of GCP services for 7.5 hours (Google Cloud Status; Google Cloud Incident Report #zKDV92iZUjeRRbpdz4h). Both were operator actions inside the provider, not external attacks or hardware failures.

Ransomware compounds the cost: **56% of downtime incidents are now cybersecurity-related** (Splunk 2024), and ransomware recovery averaged **\$2.73M per incident in 2024**, up from \$1.82M in 2023 (Sophos 2024 State of Ransomware). The 50% YoY increase is concentrated in incidents where the attacker reached the backups.

The aspiration-vs-reality split makes the gap visible: **89% multi-cloud strategy claim** (Flexera 2024), **82% say their infrastructure is not equipped for DR** (Datacore), **13% orchestrated cross-cloud DR** (Veeam 2024), **58% of servers resumed within SLA** when DR was tested (Veeam 2024), **77% of organizations that tested backups found failures** (Storage Magazine), **60% of DR plans were not effective when needed** (Alert Find).

Next thing to think about: in your last three incidents that required DR engagement, how many were caused by an operator action inside the cloud (yours or the provider's), versus an external failure?

Competitive Landscape: DR Specialists

What each tool can actually recover **to**

The single relevant question for cloud-agnostic DR is: which clouds can this tool recover **to**, regardless of where it can back up **from**?

Tool	Backup sources	Recovery targets	Structural reason
AWS Elastic Disaster Recovery	VMware, Hyper-V, physical, AWS, Azure, GCP (agent)	AWS only	Recovery orchestration is implemented as EC2 launch templates
Azure Site Recovery	VMware, Hyper-V, physical, Azure	Azure only	Replicates to Azure managed disks; no cross-cloud target
GCP Backup and DR	GCP, VMware on GCP	GCP only	Cloud-native primitive bound to GCP IAM and snapshots
Zerto (HPE)	VMware, Hyper-V, AWS, Azure	AWS + Azure	No GCP support. Block-level replication requires Zerto Virtual Manager (always-on Windows) at both ends
Druva	VMware, Azure VMs, SaaS, endpoints	AWS only	Entire Druva Cloud Platform runs exclusively on AWS infrastructure

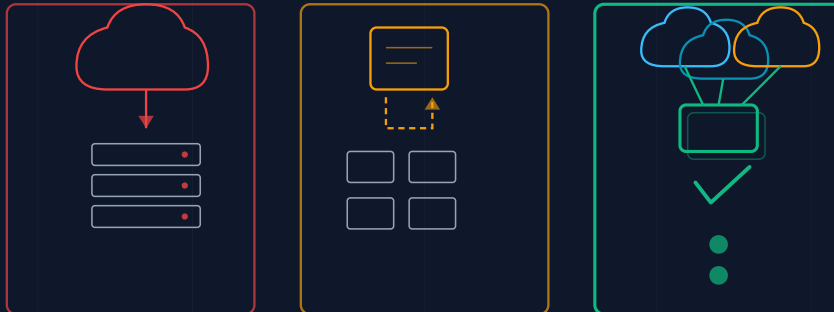
Druva is the most instructive line in the table. Its source list looks multi-cloud (it can back up Azure VMs), but every restore target is an AWS EC2 instance because the control plane and data plane are AWS-resident. A "multi-cloud backup" tool whose only recovery cloud is AWS is single-cloud DR with extra ingest paths.

Zerto comes closest to genuine portability among incumbents and is still constrained to two of three hyperscalers, with proprietary infrastructure (ZVM) required at every endpoint.

Next thing to think about: the recovery-targets column, not the backup-sources column, is what your DR plan rides on at 3am.

Competitive Landscape: Backup Vendors

Portability claims vs. portability mechanism



Veeam. Veeam's Cloud Mobility feature claims direct restore to AWS EC2, Azure, and Google Compute Engine. The mechanism: backups stored in proprietary **.vbk format**, with the Veeam software stack required at both source and target to read, convert, and instantiate. Cross-cloud restore is documented as an "efficient restore feature," not a tested, orchestrated DR capability with a recorded last-test date and achieved RTO.

The newer Veeam Data Cloud Vault is **Azure-only**, which puts a vendor-locked dependency on the path of a portability-marketed product.

Commvault. The broadest hypervisor coverage in the market: cross-hypervisor VM conversion between VMware, Hyper-V, AWS, Azure, GCP, and Nutanix. The mechanism is also the cost: every cross-cloud restore is a format conversion (VMDK to AMI to VHD to GCE), and Commvault is consistently cited in independent reviews as difficult to configure and operationally heavy. Portability exists; the cost of exercising it is what kills the runbook in practice.

Cohesity + Veritas. The December 2024 merger created the largest data protection vendor by market share, with two separate product lines, two appliance-based models, and no published cross-cloud portability roadmap as of this writing. Integration risk is the dominant near-term variable; cross-cloud "restore anywhere" is not currently in either product's shipping feature set.

The pattern across the backup tier: portability is offered as a feature on top of a proprietary backup format, with the vendor's software required at both ends to interpret the format. The dependency chain is the lock-in.

Next thing to think about: how long would it take you to read your current backups if your backup vendor went out of business tomorrow?

Vendor Lock-In: The Hidden Tax

Egress fees, proprietary formats, and the compounding bill

Three mechanisms convert "we have multi-cloud" into "we cannot actually fail over."

Egress fees. Operational data transfer between hyperscalers still runs **\$0.087 to \$0.12/GB** despite 2024 announcements waiving migration-specific egress (AWS data-transfer pricing tier 1 to 50TB; Azure bandwidth pricing zone 1; GCP general-purpose egress pricing). A cross-cloud restore of 50TB during an outage costs **\$4,350 to \$6,000 in egress alone**, before any compute or storage on the recovery side.

Mordor Intelligence puts the total TCO inflation on cross-cloud restore events at **30 to 40%**. CloudOptimo published a case in which one enterprise analytics company's egress charges grew from **\$150/month to \$2,800/month, or 25% of total cloud spend, in six months**. Egress is not friction; it is a meter that runs in proportion to how seriously you take portability.

Cloud waste compounds the problem. Flexera 2024 reports **27% of IaaS/PaaS spend is wasted**, and organizations overshoot cloud budgets by **17% on average** (Flexera 2025). A meaningful share of that waste is duplicated infrastructure maintained per-cloud precisely because data does not move freely between them.

Proprietary vendor formats are the technical complement. Veeam stores in **.vbk**. Commvault uses proprietary deduplication formats. Zerto performs block-level replication that requires Zerto-managed appliances on both sides. The backup is only as portable as the tooling required to read it, and the tooling defines the lock-in surface.

Cloud-native formats fail the same way for the same reason. EBS snapshots, Azure managed disk snapshots, and GCP persistent disk snapshots are non-portable by design: each is bound to its cloud's IAM, API, and snapshot service, and none can be mounted or read outside its origin cloud without first being exported, converted, and re-imported. The conversion is the cost.

A 10TB/month inter-cloud transfer baseline (typical for cross-cloud replication of a mid-sized production estate) is **~\$900/month in egress** before any disaster. That is what the cost of running a portable DR posture looks like as a line item; the cost of **not** running one is the \$3 economics.

Next thing to think about: model your egress bill for a 100% cross-cloud restore of your largest production cluster, then ask what fraction of that you currently budget for.

How btrfs Changes the Recovery Calculus

What makes cross-cloud restore a metadata operation



The competitive failure mode is consistent: every alternative stores backups in a format produced by, and bound to, the layer above the filesystem (hypervisor image, cloud disk snapshot, vendor backup container). Cross-cloud restore therefore requires format conversion, which is a copy plus a transform plus the vendor software stack on both ends.

Rediacc's snapshots are taken at the **btrfs filesystem layer, below the hypervisor and below the cloud API**. A btrfs snapshot is a point-in-time, copy-on-write reference to the same extents as the live filesystem: the snapshot is the data, not a serialized representation of the data. Restoring to a different cloud is a `btrfs send | btrfs receive` of those extents to a target btrfs volume.

There is no format conversion, no hypervisor matching, and no cloud-API translation; the target only needs btrfs and network. This blocks three specific failure modes:

1. **Format-conversion failures during restore.** When the failure mode is "VMDK to AMI conversion failed at 73% with no resume," the recovery clock keeps running. Filesystem-native snapshots have no conversion step to fail.
2. **Cloud-provider-level snapshot tampering.** EBS, Azure managed disk, and GCP persistent disk snapshots are mutable from the cloud's control plane: a compromise of the cloud account's IAM is a compromise of the snapshots. Btrfs snapshots are read-only at the filesystem layer; an attacker with cloud-account root cannot rewrite extents on a btrfs volume hosted on a different cloud's storage.
3. **Bandwidth blow-up on replication.** `btrfs send -p` produces incremental streams keyed on extent changes since the prior snapshot, which keeps cross-cloud replication bandwidth proportional to changed data rather than total dataset size.

How btrfs Changes the Recovery Calculus (continued)

Onboarding: what btrfs adoption actually looks like

The trade-off worth naming: this requires btrfs at both ends. Workloads on ext4, xfs, or NTFS need to be hosted on a btrfs-backed volume before the model applies.

For self-managed Linux on EC2, Azure VM, or GCE running **containerized workloads** (Docker or Compose with PostgreSQL, MySQL, MariaDB, GitLab, MinIO, Nextcloud, Jenkins, and similar), adoption is a per-volume migration to btrfs as the host filesystem on the attached block device (EBS, Azure managed disk, GCP persistent disk). The block device itself is unchanged; only the filesystem on top is reformatted. The migration shape is the standard add-volume-then-rsync-then-cutover pattern, executed once per volume, and the workload's container manifests are unmodified.

For **managed databases (RDS, Aurora, Azure SQL, Cloud SQL)** and **managed Kubernetes persistent volumes via the in-tree CSI drivers**, the host filesystem is controlled by the provider and cannot be swapped to btrfs. These workloads sit outside the model's scope today. The pattern Rediacc covers is self-managed compute with containerized databases. If a buyer's recovery-critical estate lives in RDS or Aurora, the architectural assumption of this brief does not apply to that portion of their fleet, and the cross-cloud restore argument needs to be re-evaluated against the managed service's own DR features.

Greenfield deployments adopt the model natively by provisioning btrfs at filesystem creation. There is no migration step.

The cyber-resilience corollary is the same architectural fact stated from the other side: application-layer "immutable backup" flags like S3 Object Lock governance mode can be disabled by an attacker with root in the cloud account, but a btrfs read-only snapshot hosted on a different cloud with separate credentials cannot be modified by any credential held in the origin cloud.

Next thing to think about: how much of your current restore RTO budget is consumed by format-conversion stages that a filesystem-native model would skip entirely?

Tested Failover: Proven vs. Planned Recovery

The 13% number is the real story

The DR-test statistics, taken together, describe a market in which the existence of a DR plan and the proof that the plan works have decoupled.

- **13%** of organizations use orchestrated DR workflows that actually run across environments (Veeam 2024 Data Protection Trends).
- **58%** of servers resumed within SLA during recovery tests when tests were run (Veeam 2024).
- **50%** of businesses test DR only annually or less; **7% never test at all** (Storage Magazine / Security Magazine).
- **77%** of organizations that tested backups found failures (Storage Magazine).
- **60%** of businesses say DR plans were not effective when actually needed (Alert Find).

The Veeam SureBackup feature verifies that a backup file boots and responds; it does not exercise the cross-cloud restore path. AWS DRS, Azure Site Recovery, and GCP Backup and DR each offer non-disruptive recovery tests that run inside their respective clouds, which is also where they fail when their cloud fails.

What to demand from any DR vendor, including this one: a continuously updated record of last successful cross-cloud recovery, RTO achieved on that recovery, the environments validated (which source cloud, which target cloud, which workload classes), and the script the test ran. If the vendor cannot produce that record on a per-customer basis, the DR is planned, not proven.

Next thing to think about: write down the date of your last full cross-cloud restore-and-validate. If it is more than 90 days ago, or the field is blank, the next entry in this brief applies to you.

Data Sovereignty and Cross-Cloud Snapshot Placement

Region-pinned recovery without sacrificing cross-provider failover

Readers with no data-residency constraint (US-only retail, US-only B2B SaaS without HIPAA exposure) can skip this slide; the architectural point remains useful as reference, but nothing here is load-bearing for a US-domestic deployment with no cross-border replication. For readers under GDPR, HIPAA, or financial-services data-locality rules, this slide is the placement model.

GDPR, HIPAA, and the major financial-services regulatory frameworks require that backup data physically reside in specified geographic regions and that recovery does not transit out of those regions. Most single-cloud DR tools satisfy this through region-pinned storage and region-pinned recovery, which works until the region is the failure.

Filesystem-native snapshots can be placed per-region and per-provider independently of where the workload runs. A workload in `eu-west-1` can have its primary btrfs snapshot stream targeted at an Azure West Europe storage account and a secondary stream targeted at a GCP `europe-west3` bucket. Both copies satisfy EU residency.

If AWS `eu-west-1` fails, recovery proceeds to either of the EU-resident copies without crossing the regulated boundary. Cross-provider failover and regional residency stop being a tradeoff and become two parameters of the same placement policy.

The same model handles HIPAA US-only constraints by restricting placement to US-region buckets across providers, and handles financial-services data-locality rules by pinning to single-jurisdiction copies.

Next thing to think about: if your primary region failed today, which of your residency-constrained workloads have a regulator-compliant recovery target outside that region?

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